3. Further dynamics		
3.1	Newton's laws of motion, for a particle moving in one dimension, when the applied force is variable.	The solution of the resulting equations will be consistent with the level of calculus in units P1, P2, P3 and P4. Problems may involve the law of gravitation, i.e. the inverse square law.
3.2	Simple harmonic motion.	Proof that a particle moves with simple harmonic motion in a given situation may be required (i.e. showing that $\ddot{x} = -\omega^2 x$). Geometric or calculus methods of solution will be acceptable. Students will be expected to be familiar with standard formulae, which may be quoted without proof.
3.3	Oscillations of a particle attached to the end of an elastic string or spring.	Oscillations will be in the direction of the string or spring only.